AI Pre-Production Planning and Allocation Engine for Agricultural Supply Networks in Developing Countries

Executive summary

Your concept note identifies the right economic wedge: the period **before growers and buyers lock in commitments**. In agricultural planning, that window is structurally important because first-stage decisions are made before uncertain yields and realized demand are known; one canonical fresh-produce planning paper notes that a two-stage formulation is natural in agriculture because planting-to-harvest can span roughly 10–15 weeks, while marketing contracts can set price or pricing rules before harvest. That is exactly why a pre-production decision layer is commercially meaningful and not just a modeling exercise. filecite turn0file0 [1]

The business case is viable because the loss pool is both large and concentrated in the pre-retail chain where aggregators, distributors, and processors already spend money. FAO[2] reports that 13.2 percent of food is lost after harvest and before retail, while UNEP[3]'s 2024 statistics add another 19 percent wasted at retail, food service, and household levels. The World Bank[4] separately argues that data-driven digital agriculture can reduce waste, support transparent price discovery, lower costs, and strengthen resilience. For a commercial buyer or aggregator, even modest reductions in mismatch, emergency spot buying, and poor allocation are large enough to fund software. [5]

The initial economic buyer should **not** be the smallholder farmer. It should be the party that already owns the order book, the procurement process, the contract relationships, and the logistics budget: an aggregator, distributor, buyer-led sourcing network, packhouse, processor, or exporter. IFC[6]'s smallholder-supply-chain handbook makes that logic explicit, and real-world analogs show the market is real: Twiga Foods[7] links farms to vendors and, in one development case study, was reported to pay farmers 20 to 40 percent more than brokers while settling within 48 hours; DeHaat[8] reports serving 1.8 million-plus farmers through a large center network; Ninjacart[9] positions itself around commerce, finance, and fulfillment; and AFEX[10] positions itself as a commodity trade-solutions company focused on market efficiency. The implication is that the supply side is already being digitized — but mainly around execution and market linkage, not upstream stochastic allocation. [11]

The right launch path is a **services-assisted SaaS product** that starts with historical backtesting, then runs in shadow mode, then influences actual commitment decisions. That sequencing matters because the literature on digital agricultural markets in Africa repeatedly notes a gap between pilots and scale, while IFPRI[12] has recent RCT evidence that digital-literacy support can materially raise smallholder uptake, tool usage, and trust. In practice, this means the first sale is "we will prove savings on your historical data," not "replace your planning team with AI." [13]

The most realistic near-term product is therefore a **pre-production planning and allocation engine** sold to aggregators and distributors, with farmers participating through simple messaging, field agents, or buyer-linked workflows. In a base-case illustrative model for a mid-market partner, a system that cuts mismatch-related waste by 20 percent, spot-premium spend by 35 percent, and lifts realized margin by 0.8 percent of GMV can produce roughly a 3.0x first-year value-to-cost multiple and a payback of about four months. Those figures are illustrative, but they are directionally consistent with the scale of the underlying waste and coordination problem documented by FAO and the World Bank. [5]

Why the business is viable

A pre-production planning engine is viable because it monetizes a decision that is already being made — just badly, manually, and too early. Contracts, acreage allocations, grower commitments, and buyer expectations are often fixed before the network has a reliable view of realized demand, weather-linked yields, or cross-node substitution options. Once those commitments are locked, the network's recourse is costly: emergency spot purchases, under-fill, forced markdowns, salvage, or waste. Academic work on fresh agricultural planning and contract farming points to exactly this structure: decisions happen first; uncertain demand and yield arrive later; contracts exist to reduce price risk and transaction friction, but they also constrain reallocation. [14]

That matters because this is one of the few agritech categories where the buyer has a direct, measurable P&L line. A demand-planning or agronomic-advisory tool can be important, but the budget often remains diffuse. A pre-production allocation engine, by contrast, can be attached to specific losses: avoidable waste, premium paid in emergency procurement, missed service levels, poor crop-buyer matching, and margin leakage from suboptimal commitments. Contract-farming literature notes that contracts shield farmers from spot-market volatility and help firms lower transaction costs and price uncertainty in those spot markets; a planning engine increases the value of those contracts by making the **portfolio of commitments** better before the season begins. [15]

A second reason the business is viable is that it occupies a useful middle ground between existing agritech categories. Marketplaces and B2B distribution platforms demonstrate that fragmented agricultural supply can be aggregated digitally, while ERP and inventory systems manage the world **after** commitments exist. The gap is the pre-commitment decision layer: what should be promised, to whom, from which network nodes, under what uncertainty, and with what recourse assumptions. The agri-food coordination literature explicitly notes that the field has been dominated by partial, deterministic, or two-actor analyses and that holistic coordination across actors remains underdeveloped. That gives this product a real positioning advantage if it is framed as a planning layer that complements, rather than replaces, ERP, procurement, and execution systems. [16]

Priority literature and source pack

Official and development sources should anchor the business case. The most important starting points are FAO and UNEP on food loss and waste; the World Bank on data-driven digital agriculture and digital public infrastructure in agriculture; and IFC on working with smallholders in agribusiness supply chains. Together, these sources establish the size of the economic problem, the strategic importance of smallholder-linked supply chains, and the policy trend toward shared registries, standards, API layers, and data exchange. [17]

Operations research and supply-chain sources should anchor the technical model. Ahumada and Villalobos provide a direct precedent for two-stage stochastic planning in fresh produce; Nagurney's network work is useful for multi-node supply-chain reasoning; the agri-food coordination review shows why coordination remains fragmented in the literature; contract-farming work is useful for hard constraints and risk-transfer logic; and more recent perishable-food forecasting and yield-forecasting papers help bridge the gap from academic model to deployable system. [18]

Real-world agrifood platforms should anchor the go-to-market reading of the category. Twiga, DeHaat, Ninjacart, and AFEX show there is already buyer-side willingness to digitize sourcing, aggregation, and fulfillment. The deeper lesson is not that you should copy them; it is that a planning engine can sit above, beside, or inside such networks as the intelligence layer that improves what they commit before they execute. [19]

Users, value propositions, and commercial design

The key to product design is separating **economic buyer**, **daily operator**, and **indirect beneficiary**. In most launch markets, the economic buyer will be the aggregator, distributor, processor, exporter, or organized buyer network. Daily operators will be procurement planners, supply managers, and field coordinators. Smallholder farmers are crucial users, but mostly through commitment-notification, plan-confirmation, and data-capture workflows rather than direct full-stack software administration. That matches the way agribusinesses typically work with smallholders and the way digital platforms in agricultural markets have scaled in practice. [20]

User group	Current failure mode	Product surface	Primary value proposition
Smallholder farmers	Late or unstable buyer visibility, weak price transparency, ad hoc commitments	WhatsApp/SMS/voice confirmations, agent-assisted forms, crop-plan messages	More predictable demand, better pricing discipline, fewer last-minute changes
Aggregators / FPO-linked aggregators	Over- or under-committing across farms and buyers;	Network planning dashboard, weekly risk flags, commit	Waste reduction, better grower allocation,

User group	Current failure mode	Product surface	Primary value proposition
	weak scenario visibility	recommendations	stronger service levels
Distributors / packhouses / exporters	Expensive emergency sourcing, poor fill-rate forecasting, lane imbalance	Buyer-crop-week planning board, shortage probability, recourse options	Spot-premium avoidance, margin protection, customer retention
Institutional buyers / processors / retailers	Uncertain inbound supply, low transparency before harvest	API or periodic forecast feed, confidence bands, exception alerts	Better demand-supply matching, fewer stockouts, improved sourcing confidence

The value proposition should be expressed in customer language, not research language. For the buyer-facing sale, that language is: **less waste, fewer emergency purchases, better margin, and more reliable supply**. For the farmer-facing workflow, it is: **clearer commitments, faster confirmation, less last-minute volatility, and better farm-to-buyer matching**. For a partner choosing whether to buy the system, the strongest promise is not "AI." It is: "We reduce avoidable mismatch before you lock the season." That message is also consistent with the World Bank's digital-agriculture framing around waste reduction, price discovery, lower costs, and resilience. [21]

Low-tech delivery is a requirement, not an optional "feature for later." In Sub-Saharan Africa[22], GSMA[23] estimated a 60 percent mobile-internet usage gap in 2023 and identified affordability and limited digital skills as major barriers. GSMA agritech examples such as Farmerline and Esoko emphasize voice in local languages, SMS, surveys, messaging, agent networks, and even USSD-style delivery for agricultural workflows. At the enterprise end, Meta's WhatsApp Business Platform exists specifically to let businesses communicate with customers at scale. The practical conclusion is clear: assume spreadsheet exports and simple messaging first, and richer in-app experiences only where the partner and farmer base can support them. [24]

Technical model, data, and system architecture

The core model should be a **two-stage stochastic allocation engine** over a farm–aggregator–buyer network. Stage one happens before commitments are fixed: commit volumes, buyer allocations, grower shares, reserve buffers, and allowable recourse plans. Stage two happens after uncertainty resolves: realized yields, realized demand, emergency sourcing, re-routing, salvage, and shortages. This is a direct extension of the fresh-produce stochastic planning literature into a more operational, multi-node network setting. [25]

A practical optimization objective can be written as:

$$\min_{x} C_{\text{commit}}(x) + \mathbb{E}_{s\in S}\left[C_{\text{shortage}}(x,s) + C_{\text{waste}}(x,s) + C_{\text{spot}}(x,s) + C_{\text{logistics}}(x,s) - M_{\text{grower}}(x,s)\right] + \lambda\,\text{CVaR}_{\alpha}(L_s)$$

Where x includes pre-season commitment decisions, scenario s contains realized demand and yield outcomes, and the optional CVaR term captures tail-risk exposure. The contract-farming literature is useful here because it treats demand and yield uncertainty jointly and explicitly uses risk measures when distributional knowledge is limited. [26]

The most important constraints are operational, not mathematical elegance:

- **Supply constraints:** realized farm supply equals committed base multiplied by scenario yield realizations.
- **Contract constraints:** minimums, maximums, fixed-price clauses, exclusivity, no-reallocation rules, quality grades, delivery windows, and buyer priority rules.
- **Network constraints:** lane capacities, cold-room or aggregation capacity, vehicle availability, harvest windows, lead times, and distance-time feasibility.
- **Relationship constraints:** buyer-approved farms, preferred sourcing shares, and farmer-level participation caps so the engine does not produce politically impossible recommendations.
- **Service constraints:** target fill rate by buyer, crop, and week.

The literature supports the inclusion of these features. Marketing contracts are typically set pre-harvest; contract farming reduces spot uncertainty but limits flexibility; and agri-food coordination research shows the difficulty lies precisely in actor interdependence, quality requirements, and partial visibility across the chain. [27]

Demand forecasting should be **probabilistic and hierarchical**. At a minimum, forecast weekly buyer-by-crop demand with prediction intervals, not point estimates alone. In larger partners, features should include seasonality, holidays, promotions, school or hospitality calendars, buyer-specific cadence, and price effects. In low-data contexts, use pooled models across products, zones, or buyer types so that thin series borrow strength from richer ones. More importantly, measure forecast quality using decision-relevant metrics: service-risk calibration, economically weighted error, and out-of-season robustness — not just generic accuracy. Research on perishable-food forecasting shows firms care about demand-sensing factors and performance measures aligned to waste prevention, not forecasting accuracy in isolation. [28]

Yield and supply uncertainty should be modeled at three levels. In the lowest-data setting, build empirical distributions by crop, zone, and calendar week. In the middle setting, model yield as a function of farm history, harvest window, and weather anomalies. In the highest-data setting, combine crop models, weather, and remote sensing. The yield-forecasting literature explicitly notes that low-information systems face missing management and cultivar information, which is why weather and satellite signals are so valuable. Correlation matters: some weather shocks hit multiple sourcing zones at once,

so scenarios cannot be treated as independent farm draws. OR literature on agricultural production also warns that the major difficulty is precisely the correlation structure of crop yields under weather uncertainty. [29]

Data needs and datasets to seek

A good first pilot does **not** need perfect data. It does need a disciplined minimum dataset, plus public fallback layers where partner data are weak. Publicly available sources already cover production, prices, household agriculture, rainfall, climate reanalysis, agroclimatology, and remote sensing: FAOSTAT offers broad agricultural statistics; FAO's FPMA covers domestic food prices mainly in developing countries; WFP[30]'s price database covers retail and wholesale staple-food price data; the World Bank's LSMS-ISA provides open agriculture-focused panel data in eight African countries; CHIRPS provides long-span rainfall; ECMWF[31]'s ERA5 provides hourly reanalysis with ensemble uncertainty; NASA[32] POWER provides meteorological and agroclimatology data; and ESA[33]'s Sentinel-2 provides multispectral land and vegetation imagery useful for crop monitoring. [34]

For logistics proxies in data-poor pilots, open map data and time-distance matrix APIs can approximate lane times and route distances until telematics and route-history data are available. That is not perfect, but it is usually good enough to constrain implausible allocations and estimate transport friction in a pilot-stage optimizer. [35]

Data layer	Minimum internal data to request	Public fallback or enrichment	Why it matters
Demand	12–24 months of orders by buyer, crop, week; lost-sales notes if any	FPMA, WFP price series, local market bulletins	Trains probabilistic demand model and buyer behavior
Yields / supply	Farm roster, planned acreage, historical realized yields, harvest windows	CHIRPS, ERA5, NASA POWER, Sentinel-2	Builds scenario distributions and weather-linked supply risk
Prices	Contract prices, spot prices, markdowns, salvage values	FAOSTAT price domains, FPMA, WFP	Converts operational gains into dollar savings
Contracts	Min/max volumes, exclusivity, delivery windows, penalties, payment terms	None; partner data required	Creates the real feasible set
Logistics	Node coordinates, route times, cold-storage capacity, vehicle limits	Open map data and matrix routing	Constrains network plan and recourse
Master	Farm IDs, buyer IDs, crop taxonomy, grade	Country crop taxonomies where	Prevents "bad AI" caused by basic data

Data layer	Minimum internal data to request	Public fallback or enrichment	Why it matters
data	definitions, unit conversions	available	mismatch

Commercial datasets are still worth requesting when budgets allow: buyer POS feeds, partner ERP extracts, cold-chain sensor logs, warehouse scans, telematics, agronomic field observations, and private geospatial products that improve within-season yield updates. But the launch product should assume those are optional accelerants, not prerequisites.

Required engineering and ML components

The MVP should have eight engineering and ML modules:

Module	Function	MVP requirement
Ingestion layer	Import CSV, spreadsheet, ERP extract, and messaging-derived data	Mandatory
Master-data service	Resolve farm/buyer/product IDs and units	Mandatory
Forecast service	Produce weekly probabilistic demand forecasts	Mandatory
Supply-risk service	Build yield scenarios using history plus weather inputs	Mandatory
Scenario engine	Generate correlated demand-yield scenarios and reduce them to a tractable set	Mandatory
Optimization service	Solve allocation problem under constraints and produce recommendations	Mandatory
Decision UI	Dashboard with approval workflow, not just model output	Mandatory
Messaging layer	Push confirmations/alerts via WhatsApp, SMS, or voice	Strongly recommended

Under the hood, the engineering choice should favor explainability and robustness over novelty. A relational warehouse with geospatial support, reproducible batch pipelines, probabilistic forecasting jobs, and a well-audited optimization layer will outperform a flashy black-box system that planners cannot trust.

The architecture should look like this:

```
flowchart LR
    A[Buyer orders and contract signals] --> B[Demand forecasting]
    C[Farm roster and yield history] --> D[Supply-risk modeling]
    E[Weather and satellite data] --> D
    F[Prices, logistics, capacities] --> G[Scenario generator]
```

```
B --> G
D --> G
G --> H[Two-stage stochastic optimizer]
H --> I[Planner dashboard]
H --> J[Farmer and field-agent messaging]
I --> K[Approve / edit commitments]
J --> K
K --> L[Committed plan]
L --> M[Execution and realized outcomes]
M --> N[Backtest and model monitoring]
N --> B
N --> D
```

This architecture is the practical expression of the literature: upstream information is fused into scenario-based decisions, then the actuals are fed back for continuous improvement. [36]

Pilot, backtesting, and measurement

The pilot should be designed as an **evidence program**, not a software demo. The literature on digital agricultural markets in Africa is unusually clear on this point: many pilots fail to scale because they do not generate credible evidence, do not fit operational workflows, or stop at the "interesting idea" stage. Separate work on perishable-food forecasting also shows firms want waste-relevant KPIs and development pathways, not just algorithmic novelty. [37]

A strong pilot has three layers of proof.

First, run a **historical replay backtest**. For each historical planning date, restrict the model to information that would genuinely have been available at that date, generate demand and yield scenarios, solve the allocation, and compare the counterfactual optimized plan to the actual historical plan under the same realized outcomes. This removes hindsight bias.

Second, run the tool in **shadow mode** with at least two independent partners or two materially separate networks inside one group. The planning team keeps making real decisions, but the system logs what it would have recommended and how those recommendations would have changed expected waste, shortfall risk, and spot exposure.

Third, run a **limited live pilot** on selected crop-buyer lanes once the partner is comfortable with the shadow-mode output.

Recommended backtesting design

Backtest element	Recommended design
Decision unit	Buyer × crop × week × sourcing zone
Historical window	Minimum 52 weeks; preferably 104+ weeks covering two seasons

Backtest element	Recommended design
Forecast protocol	Rolling-origin evaluation with only contemporaneously available features
Counterfactuals	Actual historical plan; simple planner heuristic; optimized plan; hindsight upper bound
Scenario design	Demand scenarios + correlated yield scenarios + recourse cost assumptions
Primary outcomes	Waste cost, spot-premium spend, fill rate, gross margin, shortage probability
Secondary outcomes	Grower fulfillment stability, buyer service consistency, planner overrides
Acceptance rule	Positive economics in at least two partners and one live-lane cohort

Statistical methods that are appropriate

For the backtest, use **paired comparisons** because the relevant question is whether the optimized decision outperforms the actual decision on the same realized history. Because agricultural demand and supply have serial dependence, estimate uncertainty with **block bootstrap confidence intervals** over planning windows rather than naively treating weeks as iid observations. For live pilots where treated and untreated lanes coexist, use **difference-in-differences** or matched-lane comparisons to isolate effect from seasonality, weather shocks, and buyer-specific drift. For forecast-model selection, compare probabilistic forecasts using calibrated interval coverage, pinball loss, and decision-cost outcomes, not accuracy alone.

KPIs and financial impact metrics

KPI	Definition
Waste reduction %	$(\text{Baseline waste} - \text{Pilot waste})/\text{Baseline waste}$
Spot-premium dollars saved	Baseline emergency-buy quantity × premium minus pilot equivalent
Fill-rate uplift	Pilot OTIF or service level minus baseline
Gross-margin uplift	Pilot gross margin minus baseline gross margin, net of implementation cost if desired
Commitment accuracy	Realized fulfilled demand / committed supply within defined tolerance band
Shortfall probability	Share of planning intervals with under-fill under scenario distribution
Planner adoption	Share of recommendations accepted or edited rather than ignored
ROI	Annualized customer benefit / first-year software and

KPI	Definition
	implementation cost
Payback	First-year cost divided by monthly run-rate savings

Pilot partner profiles and expected outcomes

The table below is intentionally **illustrative**, not empirical. It shows the kinds of partners most suitable for the first wave.

Pilot partner profile	Why it is attractive	Minimum data readiness	Expected outcome range
Urban fresh-produce distributor	Pain is visible in daily stockouts, markdowns, waste, and rush procurement	Medium: weekly orders, supplier roster, route and waste logs	Waste cost down 15–25%; spot-premium spend down 25–40%; payback in one season
Aggregator or FPO-linked sourcing network	Strong smallholder relevance and direct farmer-allocation problem	Low to medium: farm roster, buyer commitments, basic seasonal history	Over-commitment down 20–35%; buyer fill rate up 3–6 points; farmer-side allocation stability up materially
Processor- or exporter-linked grower network	Contract logic is clearer and quality constraints are explicit	Medium to high: contract terms, specification grades, procurement records	Emergency sourcing down 20–45%; contract fill rate up 4–8 points; realized margin up through better crop-buyer matching

Recommended visualizations

Visualization	What it should show	Why it matters
Supply-demand lock-in timeline	Weekly committed supply against P10/P50/P90 demand envelopes	Makes the pre-commitment risk visible before harvest
Shortfall heat map	Crop × buyer × week shortage probability	Directs planner attention to the highest-risk lanes
Waste and spot waterfall	Baseline economics versus optimized economics	Converts model output into CFO language
Network risk map	Farm, hub, and buyer nodes with color-coded risk	Shows concentration risk and lane fragility
Forecast calibration chart	Predicted bands versus realized volumes	Builds trust in the probabilistic layer
Planner override	Accepted, edited, and rejected	Reveals adoption friction and

Visualization	What it should show	Why it matters
dashboard	recommendations by user	model-governance issues

Go-to-market, pricing, organization, and launch roadmap

The most defensible go-to-market is **aggregator-first, buyer-led, and services-assisted**. That fits the way agribusinesses already source from fragmented smallholders, and it avoids the hardest monetization problem in agritech, which is trying to charge individual farmers for upstream supply-chain optimization. Real-world platforms have already shown that commercial value accrues where demand aggregation, fulfillment, and payment orchestration sit. The planning engine should therefore be sold as a profit-protection and service-level tool into the organizations already coordinating those flows. [38]

Pricing should begin simple. The best early structure is a **paid pilot plus annual software contract**, with an optional performance bonus after results are validated. A practical model is:

- **Discovery and backtest fee:** one-time fee for data audit, cleaning, scenario design, and backtest.
- **Annual platform fee:** priced by network complexity, number of crops, planning frequency, and number of nodes.
- **Outcome-based component:** optional share of validated savings after trust is established.

Pure outcome-based pricing sounds attractive, but early on it complicates procurement, attribution, and customer trust. A hybrid model is easier to close.

Organizational roles and partnerships

Role	Primary responsibility
Product and commercial lead	Owns customer design, pricing, pilot scope, and procurement
Optimization lead	Owns model formulation, constraints, solver strategy, and scenario logic
Forecasting lead	Owns demand and yield models, calibration, and monitoring
Data engineer	Owns ingestion, master data, pipelines, and auditability
Full-stack engineer	Owns dashboard, approvals, workflow, and integrations
Implementation manager	Owns partner onboarding, data requests, training, and weekly review cadence
Field operations liaison	Owns farmer messaging, agent processes, and exception handling
Legal / data governance support	Owns consent language, sharing terms, and contract protections

The partner set should include at least one messaging provider, one or more pilot agribusinesses, field-agent or FPO distribution support where needed, and an applied OR or supply-chain advisor who can help pressure-test assumptions but not overcomplicate the MVP. The World Bank's recent work on agriculture-oriented digital public infrastructure and data labs is helpful here: it points toward shared registries, common data standards, and public-private experimentation rather than one-off bespoke systems. [39]

MVP and pilot timeline

An MVP can be built in roughly nine months if scope is controlled. A realistic calendar beginning in **May 2026** looks like this:

```
gantt
    title MVP and Pilot Roadmap
    dateFormat  YYYY-MM-DD
    axisFormat  %b %Y

    section Discovery
    Partner selection and data request          :a1, 2026-05-11, 2026-05-29
    Data audit and schema mapping               :a2, 2026-05-25, 2026-06-19

    section MVP Build
    Baseline forecast and scenario engine       :b1, 2026-06-15, 2026-07-24
    Optimization MVP and solver tuning          :b2, 2026-07-06, 2026-08-14
    Dashboard and messaging workflows           :b3, 2026-08-03, 2026-09-04

    section Evidence
    Historical backtesting with two partners    :c1, 2026-08-17, 2026-09-25
    Shadow mode planning                        :c2, 2026-09-28, 2026-10-30

    section Live Pilot
    Controlled live deployment                  :d1, 2026-11-02, 2026-12-18
    Pilot closeout and commercial conversion    :d2, 2027-01-04, 2027-01-29
```

The gates should be explicit. Do not enter shadow mode until the backtest is reproducible. Do not enter live mode until planner overrides are explainable. Do not sell multiyear enterprise contracts until at least one customer has a usable ROI narrative backed by auditable calculations.

Financial model and sensitivity

The two tables below are **illustrative modeling assumptions**, not market facts. They are intended to show what a plausible customer business case might look like for a mid-market distributor or aggregator with meaningful pre-retail mismatch costs.

Illustrative customer assumptions

Assumption	Value

Assumption	Value
Annual GMV	$10,000,000
Baseline mismatch-related waste cost	3.0% of GMV
Baseline spot-premium spend	1.8% of GMV
Annual platform fee	$48,000
One-time implementation and onboarding	$20,000
First-year customer cost	$68,000

Illustrative scenario outcomes

Scenario	Waste reduction	Spot-premium reduction	Margin uplift as % of GMV	Waste savings	Spot savings	Margin uplift	Total annual value	First-year cost	Net benefit	Value / cost	Payback months
Conservative	10.0%	20.0%	0.5%	$30,000	$36,000	$50,000	$116,000	$68,000	$48,000	1.71x	7.0
Base	20.0%	35.0%	0.8%	$60,000	$63,000	$80,000	$203,000	$68,000	$135,000	2.99x	4.0
Upside	30.0%	50.0%	1.2%	$90,000	$90,000	$120,000	$300,000	$68,000	$232,000	4.41x	2.7

Sensitivity of value-to-cost multiple

The next table shows how the economics move if only waste reduction and spot-premium reduction change, while holding margin uplift at a modest 0.6 percent of GMV.

Waste reduction ↓ / Spot-premium reduction →	20%	35%	50%
10%	0.85x	1.25x	1.65x
20%	1.29x	1.69x	2.09x
30%	1.74x	2.13x	2.53x

The financial reading is straightforward. This product does not need dramatic algorithmic superiority to work. It needs **modest but repeatable operational improvement** in a cost base that is already significant. That is why the sale should start where waste, spot buys, and service failures are already visible in management reporting. That framing is also consistent with the official policy and research literature emphasizing waste reduction, lower transaction costs, better market matching, and resilience rather than “AI” as an end in itself. [40]

Risks, scaling, and execution prompt

The non-technical risks are as important as the model.

Risk	Why it matters	Mitigation
Weak master data	Bad IDs and unit mismatches can destroy trust faster than a mediocre model	Start with schema mapping, unit normalization, and an auditable data dictionary
Low trust and digital friction	Adoption fails when planners and farmers do not understand or trust recommendations	Human-in-the-loop approvals, field-agent training, and clear exception handling
Data-rights ambiguity	Agricultural data is governed by contract law, competition law, privacy law, and IP concerns	Explicit data-sharing clauses, consent design, role-based access, and export rights
Correlated weather shocks	Multi-zone failures can invalidate simplistic diversification assumptions	Use weather-zone scenario generation and tail-risk metrics
Side-selling or contract non-performance	Operational reality can diverge from model assumptions	Model feasible rather than ideal behavior; track performance by node
Pilot-to-scale gap	Many pilots remain experiments rather than operating systems	Pre-register metrics, prove economics, and design for existing workflows first
Overbuilt product	Teams often overinvest in dashboards before evidence and workflow fit	Build backtest and shadow-mode evidence before adding enterprise complexity

The policy and institutional literature strongly supports taking governance seriously. FAO's digital-agriculture work explicitly highlights regulatory frameworks, data governance, and farmer rights; World Bank work notes that agricultural-data use is shaped by contract, competition, privacy, and IP law; and IFPRI's recent work emphasizes that digital adoption depends heavily on training, usability, and trust. [41]

Scaling should proceed in layers. Start with one region, a small set of crops, and one planning cadence. Then add more buyers, more sourcing zones, and more sophisticated scenario generation. Over time, the data moat becomes meaningful: better farm-level baselines, better buyer segmentation, richer route-time priors, and stronger calibration of yield-demand dependence. The World Bank's recent agriculture-DPI and roadmap work is especially relevant here because it points toward common registries, API-compliant data exchange, market-discovery services, and public-private data labs that make scaling easier than bespoke integrations alone. [39]

Use the following Claude Code prompt to turn this report into an execution plan:

You are Claude Code acting as a product, data, and pilot-operations lead for an AI-powered pre-production planning and allocation engine for agricultural supply networks in developing countries.

Objective:
Create a concrete 90-day execution package for an MVP and pilot program for a decision-layer system that helps farms, aggregators, distributors, and buyers make better pre-production commitments before yield and demand are fully known.

Core product concept:
- Two-stage stochastic allocation under demand uncertainty, yield uncertainty, weather risk, logistics constraints, and contract constraints
- Initial customer is an aggregator/distributor/buyer network, not individual farmers
- Farmer interactions must support low-tech channels: WhatsApp, SMS, voice, field-agent-assisted workflows
- Product should begin with historical backtesting, then shadow mode, then controlled live pilot
- Assume data can be sparse, messy, and partially reconstructed from spreadsheets, invoices, and chat exports

Your tasks:
Produce a structured output with the following sections and artifacts.

Section A: data request package
1. Create a pilot data request checklist for two partner types:
   - distributor / aggregator
   - buyer-led contract sourcing network
2. For each dataset, specify:
   - business owner
   - file format
   - grain
   - minimum fields
   - ideal fields
   - frequency
   - common data-quality issues
3. Provide CSV schemas for:
   - farms.csv
   - buyers.csv
   - products.csv
   - contracts.csv
   - orders.csv
   - yields.csv
   - logistics_lanes.csv
   - spot_purchases.csv
   - waste_events.csv
4. List which public datasets can be used as fallback enrichments if partner data is missing:

```
   - prices
   - weather
   - rainfall
   - remote sensing
   - logistics and routing

Section B: repository and code plan
1. Propose a repo structure for the MVP:
   - data ingestion
   - cleaning
   - feature engineering
   - forecasting
   - scenario generation
   - optimization
   - evaluation
   - dashboards
   - docs
2. Create pseudocode and file-by-file responsibilities for:
   - ingest_partner_data.py
   - normalize_master_data.py
   - build_demand_features.py
   - train_probabilistic_demand_model.py
   - build_yield_scenarios.py
   - generate_correlated_scenarios.py
   - solve_two_stage_allocation.py
   - replay_backtest.py
   - evaluate_kpis.py
   - export_dashboard_tables.py
3. Include the optimization objective, main decision variables, and key
constraints in implementation-ready notation
4. Add notes on where human overrides and audit logs must be captured

Section C: experiment and pilot design
1. Write a detailed backtesting protocol:
   - rolling-origin setup
   - data leakage controls
   - baseline heuristics to compare against
   - perfect-information upper bound
2. Define statistical methods:
   - paired comparisons
   - block bootstrap confidence intervals
   - difference-in-differences for live pilots
3. Define primary and secondary KPIs with formulas:
   - waste reduction %
   - spot-premium dollars saved
   - gross-margin uplift
   - fill-rate uplift
   - shortfall probability
   - planner adoption / override rate
   - ROI and payback
```

4. Build a 12-week shadow-mode operating cadence:
   - weekly planning run
   - recommendation review
   - override capture
   - actuals reconciliation
   - executive readout

Section D: stakeholder communication assets
Create templates for:
1. Pilot partner outreach email
2. Data access and security request email
3. Weekly pilot steering-committee update
4. Farmer-facing message templates for:
   - plan confirmation
   - expected delivery window
   - quantity revision alert
   - exception / risk alert
5. Internal one-page memo for procurement, operations, and finance leaders explaining:
   - what the system does
   - what it does not automate
   - why it should be trusted
   - how success will be measured

Section E: product and dashboard specification
1. Define the MVP screens:
   - network overview
   - shortfall / overage risk board
   - crop-buyer-week allocation table
   - scenario comparison view
   - planner override log
   - pilot KPI dashboard
2. For each screen, specify:
   - target user
   - interaction goal
   - required inputs
   - outputs
   - warning states
3. Suggest visualizations:
   - supply-demand lock-in timeline
   - waterfall of economics
   - heat map of shortfall risk
   - network risk map
   - calibration chart
4. Include Mermaid diagrams for:
   - system architecture
   - planning workflow
   - pilot timeline

```
Section F: operating model and staffing
1. Recommend the minimum team for the first 9 months:
   - product/commercial
   - optimization
   - forecasting/data science
   - data engineering
   - full-stack engineering
   - implementation/field operations
2. Define responsibilities, hiring order, and what can be outsourced
3. Identify external partnerships that are required versus optional

Section G: deliverables and milestone plan
1. Convert all of the above into a dated 9-month workplan
2. Include milestones, dependencies, and go/no-go criteria for:
   - discovery complete
   - data readiness complete
   - backtest complete
   - shadow mode ready
   - live pilot ready
   - commercial conversion ready
3. Add a risk register with probability, impact, owner, and mitigation
4. End with a concise executive checklist of the next 10 actions

Constraints:
- Do not ask me clarifying questions unless absolutely blocked; make explicit
assumptions instead
- Optimize for low-tech and data-poor environments
- Prefer simple, explainable models and workflows over complexity
- Assume farmers may interact through agents and messaging, not a rich native
app
- Format the output in clean markdown with tables, checklists, and Mermaid
blocks where useful
- Be concrete enough that an engineer and an implementation manager could
start work immediately
```

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